

Case Study 1 - William Armentrout

Introduction:

In recent years, a large amount of attention has been spent on billionaires with regard to their wealth. Particularly with extra scrutiny looking into what I will be describing as 'tech billionaires' who are at the forefront of investments in Artificial Intelligence.

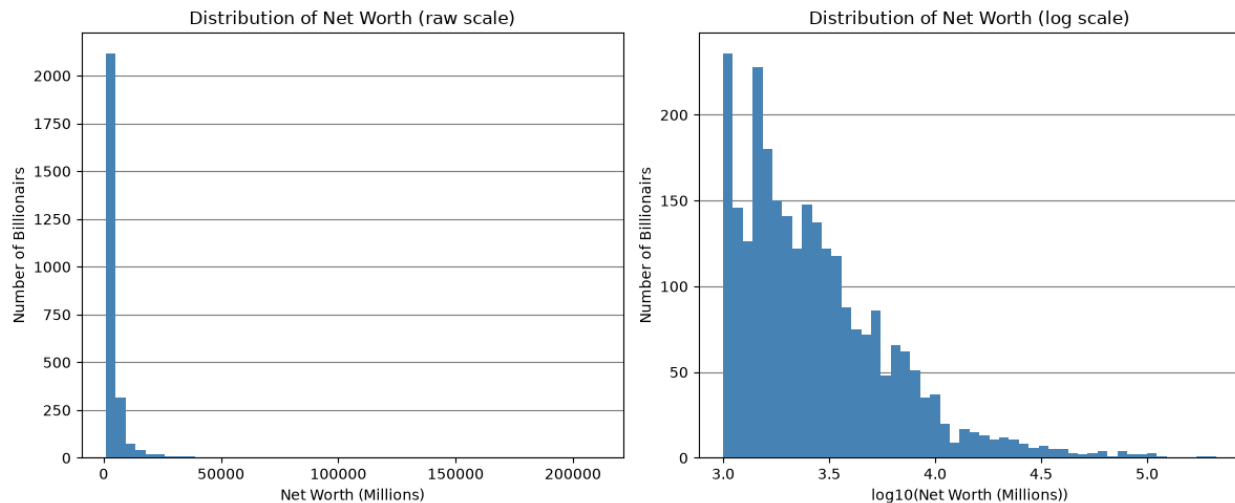
This case study is looking into three crucial research questions regarding billionaires' wealth and potential correlations between them, their industries, and their country of origin.

- R1: Do billionaires in the technology industry have higher average net worth than those in other industries?
- R2: Is there a relationship between age and net worth among billionaires?
- R3: Does the number of Billionaires per country correlate with GDP size?

Here are the hypotheses statements and null hypotheses I plan to use to test the research questions above:

- H1: Billionaires in the tech industry have a significantly higher average net worth than billionaires in other industries.
 - H0: There is no significant difference in average net worth between tech and non-tech billionaires.
- H2: There is a statistically significant relationship between a billionaire's age and their net worth.
 - H0: Age and net worth are not significantly related.
- H3: The number of billionaires in a country is positively correlated with that country's GDP.
 - H0: There is no positive correlation between a country's billionaire count and its GDP.

Before we get into evaluating the research questions, we need a brief overview of the Billionaires.csv dataset. In order to determine which tests to use, we need to know the normality of the billionaires' wealth distribution. Below is a histogram of the distribution of net worth in millions of dollars. I used both the raw scale and the log scale due to the difference in magnitudes of wealth to capture the data better.



As you can see in the histograms above, the dataset is heavily skewed, meaning we can't assume normality. This will need to be taken into account in the methods used to evaluate the research questions.

Methods / Results:

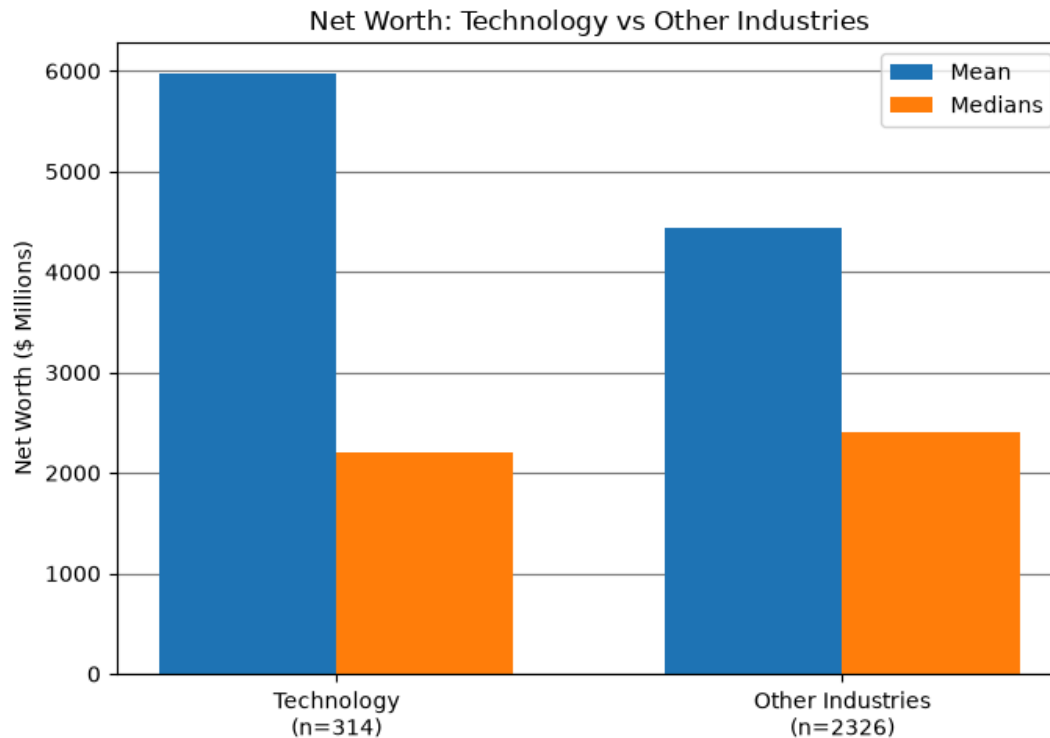
Research Question 1:

Do billionaires in the technology industry have higher average net worth than those in other industries?

For the first research question, I used the mean and median to compare the wealth of the two groups at face value, along with both the Welch Independent-Samples t-test and the Mann-Whitney U test. The t-test and U-test were both used due to how skewed the histogram was. The t-test assumes normality, so its results would be questionable even with a low p-value. Here are the results from the relevant code block:

```
Technology: n=314, mean=$5,980.6M, median=$2,200.0M
Other:      n=2326, mean=$4,440.6M, median=$2,400.0M
Welch's t-test: t=1.891, p=0.0595
Mann-Whitney U: U=369392.5, p=0.7397
```

Given the results above, the mean is higher for tech, but the median is slightly lower than the other industries. Also, neither the t or U test was significant, as both had p-values exceeding 0.05. This is likely due to extreme tech outliers such as Musk and Bezos inflating the mean for tech billionaires. This indicates that tech billionaires are not inherently wealthier than those in other industries. Below is a bar chart to better visualize the difference between tech and other industries' billionaires.



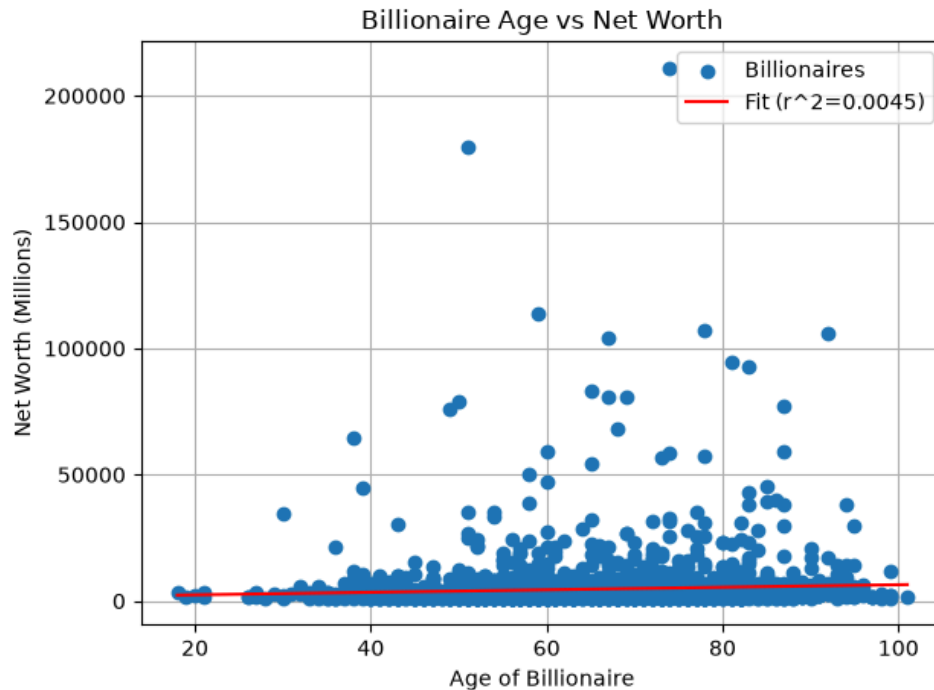
Research Question 2:

Is there a relationship between age and net worth among billionaires?

For the second research question, I used both the Pearson and Spearman correlation coefficients along with a simple linear regression. Spearman's was included once again because Pearson's coefficient is not as resistant to outliers as Spearman's. Here are the results from the relevant code block:

```
Pearson r = 0.0671, p = 0.0007
Spearman rho = 0.1440, p = 0.0000
Regression: slope=50.28, R^2 = 0.0045, p = 0.0007
```

Given the results above, we can see that there is, in fact, a statistically significant positive correlation between the age of a billionaire and their net worth, as p-values for Pearson, Spearman, and Regression are all below 0.05. However, it is important to note that the slope of ~\$50M may seem large initially, but on the scale of Billionaires network is relatively small, especially considering there are some with a net worth exceeding \$100,000M or \$100B. Below is a graph containing the scatter plot of billionaires' age and their net worth:



Research Question 3:

Does the number of Billionaires per country correlate with GDP size?

For the third and final question, I used a country-level aggregation and then Pearson's correlation coefficient on both the raw and log values of both GDP and billionaire counts since they span orders of magnitude across various countries. I also completed a linear regression on the raw and log data. Here are the results from the relevant code block:

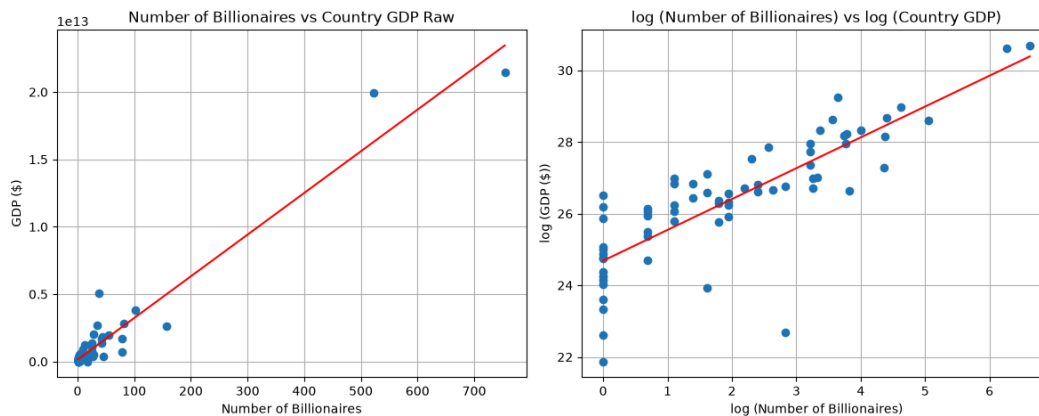
```
Pearson r = 0.9702, p = 0.0000
Pearson r (log-log) = 0.8200, p = 0.0000
Regression: slope=30913831052.91, R^2 = 0.9412, p = 0.0000
Regression (log-log): slope=0.86, R^2 = 0.6724, p = 0.0000
```

When looking at the raw Pearson r score, we see a strong linear relationship, but looking at the left scatter plot below, we see two outliers with extremely high GDP compared to the other countries. Those two points have a large amount of leverage, so much that the regression line is basically designing itself around them, making the R^2 value look stronger than it should. So the 0.94 from the raw regression is overinflated.

To get a better representation of the data, we transform both variables with a log function. When looking at the log scatter plot, we can see that for countries with $\log(\text{billionaire count}) \approx 0$ or 1 billionaire, $\log \text{GDP}$ varies greatly from 22 to 27, which is orders of magnitude in variation. This indicates that among countries with few billionaires, GDP is not well predicted by billionaire count

alone. The results indicate that while a relationship is real, it is a lot noisier than the original R^2 value might have suggested.

The slope from the log regression is 0.86, indicating that a 1% increase in the number of billionaires is associated with a 0.86% increase in GDP. This could indicate that billionaire count grows somewhat faster than GDP does in economies of scale or that added billionaires are associated with less than proportional GDP gains.



Discussion

Hypothesis 1: fail to reject the null, as the p-value of the U test is 0.7397, vastly over the 0.05 needed for significance. This is supported by the mean/median disagreement, which suggests extreme outliers, such as Musk and Bezos, drive the tech-wealth billionaire narrative over other types of billionaires. Hypothesis 2: Rejects the null hypothesis, indicating there is a relationship between billionaires' net worth and age. However, an incredibly small R^2 value of 0.0045 and a regression slope that is small for this magnitude of wealth indicate that the relationship is likely trivial in any practical terms despite the significance of the correlation. Hypothesis 3: Rejects the null hypothesis, indicating a relationship between the number of billionaires and GDP. However, in the raw results, the R^2 value of 0.9412 is heavily overinflated by the presence of two extreme outliers, and a more accurate measurement would be that of the log regression with an R^2 of 0.6724, which still shows the same relationship, but it is not as drastic as first shown with the raw data.

There are a few limitations with the dataset. The first is missing population metrics; this is important because countries with a larger population are going to have a larger GDP and number of billionaires, which may have implications for H3. The second limitation is survivorship bias. The dataset only includes those who have already crossed the billion-dollar threshold, without a comparison group of those near a billion or those who lost their wealth; it limits any claims about what drives someone into this wealth bracket.